Predicting ICLR 2025 Peer-Review Decisions with Bayesian Models

CSE 5825 Bayesian Machine Learning — Jiazheng Li

September 2026

Research question and data

Can information in peer reviews predict whether an ICLR 2025 submission is rejected or accepted as a poster, spotlight, or oral presentation? More specifically, do reviewer disagreement, changes after author rebuttal, and discussion participation contain information beyond the mean review rating?

The proposed dataset combines two public, OpenReview-derived releases for ICLR 2025. The ICLR Insights score-change data identify each paper by its OpenReview ID and provide each reviewer's initial and post-rebuttal rating, confidence, and other rubric scores. The corresponding Paper Copilot OpenReview archive provides paper metadata, final decision status, research area, and reply-count fields such as `reply_authors` and `reply_reviewers`. The unit of analysis will be a paper: reviewer-level scores are aggregated within each paper before predicting its decision. We will join records by OpenReview paper ID, retain papers with a known final decision and usable reviews, and treat withdrawals and desk rejections separately because they do not follow the ordinary reviewed-paper path. The four target labels are *Reject*, *Poster*, *Spotlight*, and *Oral*. Focusing on one year keeps the review rubric and conference decision policy consistent.

Table 1 describes the complete ICLR 2025 submission pool in an independent OpenReview export. It gives the scale and class imbalance of the target outcome; these are *not yet* the class counts in our joined analysis sample. Among the 11,672 submissions, 11,520 have review records in that export. In particular, oral and spotlight decisions are rare, so their predictive performance will need separate evaluation.

Final decision	Papers	Share of all submissions
Reject	7,893	67.62%
Poster	3,115	26.69%
Spotlight	380	3.26%
Oral	213	1.82%
Desk reject	71	0.61%
Total	11,672	100%

Table 1: ICLR 2025 decision counts in the full submission pool, before our dataset join and exclusions. Source: ReviewerToo ICLR 2025 full dataset card; percentages calculated from its counts.

The ICLR Insights study reports 46,748 reviews for these 11,672 papers, of which 46,353 have a paired initial/final score record. It reports paper-level score trajectories for 11,431 papers (Table 2); some initial scores were not captured, so the paired subset is smaller than the submission pool. An ICLR 2025 overall rating is one of $\{1, 3, 5, 6, 8, 10\}$, rather than any integer from 1 to 10. The mean paper rating rises from 4.78 before rebuttal to 5.15 afterward in the paired subset. The percentages

in Table 2 are descriptive: papers with rising scores also start and finish at different rating levels from papers with unchanged scores. Thus, the table does not test H2's comparison *at the same final mean rating*.

Change in paper mean	Papers (share)	Mean rating: initial → final	Accepted
Increase	5,807 (50.8%)	5.21 → 5.97	55.7%
No change	5,247 (45.9%)	4.30 → 4.30	7.8%
Decrease	377 (3.3%)	4.88 → 4.41	8.0%
All paired papers	11,431 (100%)	4.78 → 5.15	32.1%

Table 2: Published paper-level ICLR 2025 score-change statistics, before joining to reply fields. Counts, rating means, and acceptance rates are from Table 2 of Kargaran et al.; shares calculated from the reported counts.

The score-change release stores initial and final reviewer scores as semicolon-separated pairs and provides a `tracing_score` for confidence in matching a reviewer across snapshots. We will check this score when building trajectories and report the final join size and missingness by decision class. An audit of the Paper Copilot final and January 22 snapshots found that their normalized `reply_authors` and `reply_reviewers` fields are zero throughout the records examined. We therefore cannot treat those fields as measured discussion participation. H3 and the discussion predictor require separately reconstructed, timestamped OpenReview thread events; without them, we will fit the rating-based model and report H3 as untested. We will also reconcile decision labels across releases before forming the analysis sample.

Variables and probabilistic view

For paper i and reviewer j, the recorded variables are the initial and post-rebuttal ratings $r_{ij}^{(0)}$ and $r_{ij}^{(1)}$, the corresponding confidence scores $c_{ij}^{(0)}$ and $c_{ij}^{(1)}$, the research area a_i, the number of reviewers J_i, and the final decision Y_i. We summarize ratings by their mean $\bar{r}_i^{(t)}$ and standard deviation $D_i^{(t)}$ at time $t \in \{0, 1\}$. The change in mean rating is $\Delta_i = \bar{r}_i^{(1)} - \bar{r}_i^{(0)}$. If reliable thread events become available, let U_i and V_i indicate a pre-decision author reply and a subsequent reviewer reply, respectively, and set $B_i = U_i V_i$. These are planned observed variables, not values supplied reliably by the current normalized reply-count fields.

This graph expands $p(Y_i \mid \mathbf{x}_i, \theta)$ into acceptance and presentation-tier stages. Initial ratings may be associated with whether authors reply; the model does not assume the components of $\mathbf{x}_i$ are independent. It conditions on their observed joint values, without fitting $p(\mathbf{x}_i)$ or a paper-specific latent quality variable. The unknown shared parameters comprise acceptance coefficients $\boldsymbol{\beta}_A$, area effects u_a, and tier coefficients W. The initial mean $\bar{r}_i^{(0)}$ and change Δ_i enter the regression; the final mean is used to compute Δ_i, avoiding three exactly dependent linear predictors.

Let $A_i = 1$ if paper i is accepted and $A_i = 0$ if it is rejected. A concrete after-rebuttal *likelihood* is

$$A_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \alpha + \beta_0 \bar{r}_i^{(0)} + \beta_D D_i^{(0)} + \beta_c \bar{c}_i^{(0)} + \beta_J J_i + \beta_\Delta \Delta_i + \beta_B B_i + \beta_{0B} \bar{r}_i^{(0)} B_i + u_{a(i)}.$$

Here p_i is the acceptance probability, $\bar{c}_i^{(0)}$ is mean initial confidence, and $u_{a(i)}$ is the unknown effect of research area a_i. The interaction permits the predictive association of replies to differ by initial rating. We will include terms involving B_i only if timestamped replies can be verified. H2 is

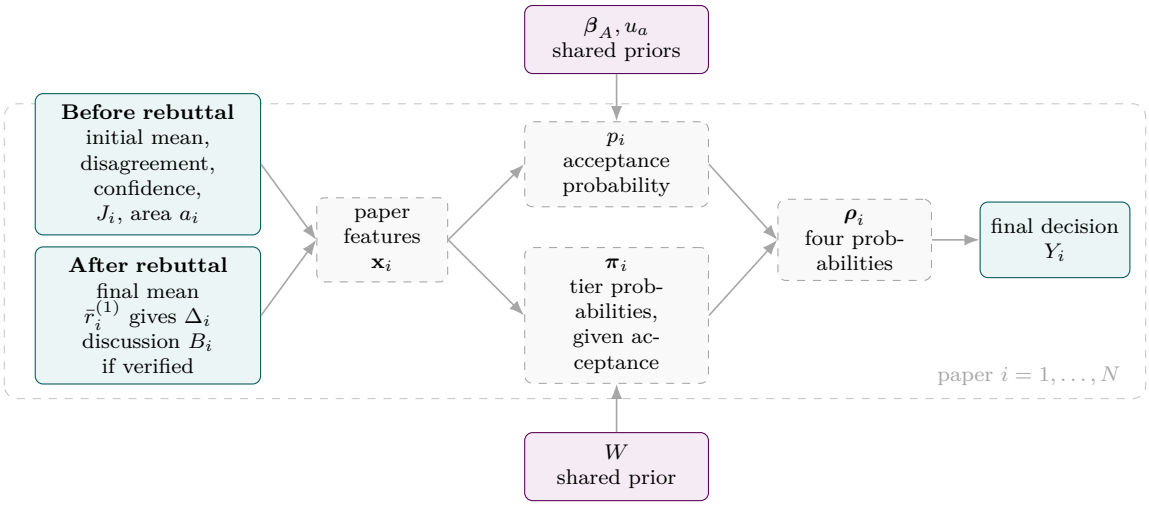


Figure 1: Two-stage conditional graphical model. Teal nodes are observed inputs or outcomes, purple nodes are shared unknown parameters with priors, and dashed nodes are deterministic calculations. The plate repeats for each paper. The model conditions on observed features; it does not model their joint distribution. The discussion feature B_i enters only if verified thread events are available.

evaluated by comparing posterior predictions at a fixed final mean while changing Δ_i and setting $\bar{r}_i^{(0)} = \bar{r}_i^{(1)} - \Delta_i$ accordingly. H1 and H3 likewise use posterior predictive comparisons at the stated matched initial ratings rather than treating any single coefficient as the entire hypothesis test.

Among accepted papers, let $T_i \in \{P, S, O\}$ denote Poster, Spotlight, or Oral. A second regression gives

$$T_i \mid A_i = 1, \mathbf{x}_i, W \sim \text{Categorical}(\boldsymbol{\pi}_i), \qquad \boldsymbol{\pi}_i = \text{softmax}(W\mathbf{z}_i),$$

where $\mathbf{z}_i$ encodes the observed predictors and a constant, with the Poster row of W fixed to zero for identification. The two stages produce

$$\boldsymbol{\rho}_i = (1 - p_i,\ p_i\pi_{i,P},\ p_i\pi_{i,S},\ p_i\pi_{i,O}), \qquad Y_i \mid \mathbf{x}_i, \theta \sim \text{Categorical}(\boldsymbol{\rho}_i).$$

After standardizing the continuous predictors, one possible set of *priors* is

$$\beta_\ell \sim \mathcal{N}(0,1), \quad W_{k\ell} \sim \mathcal{N}(0,1)\ (k \in \{S,O\}), \qquad \alpha \sim \mathcal{N}(0, 2^2), \qquad u_a \sim \mathcal{N}(0, \tau^2), \quad \tau \sim \text{HalfNormal}(1).$$

These independent coefficient priors are a convenient starting choice; they do *not* assert that the observed predictors are independent. The shared prior on u_a lets areas with little data borrow information from other areas. The acceptance stage learns from every labeled training paper, whereas the tier stage learns from accepted training papers. Updating the priors gives $p(\theta \mid \mathcal{D}_{\text{train}})$; averaging $\boldsymbol{\rho}_i$ over posterior draws gives four predictive probabilities for a held-out paper. For an *initial-review* prediction, we will fit a separate version using only initial ratings and confidence; rebuttal changes and replies are unavailable at that time. Reviewer IDs are local to each paper, so we will not try to estimate persistent reviewer strictness across papers.

Three initial hypotheses

H1. Reviewer disagreement. Among papers with similar *initial* mean ratings, greater dispersion among reviewers is associated with a higher probability of rejection. This asks whether

the distribution of ratings matters beyond their average.

H2. Rebuttal trajectory. Among papers with similar *final* mean ratings, papers whose ratings rose from a lower initial mean have a different probability of acceptance or presentation tier from papers whose ratings remained stable. The direction of this difference is left open because the initial concern and its resolution may carry competing signals.

H3. Discussion participation. Among papers with similar initial ratings and disagreement, papers with both author and reviewer replies during rebuttal have a different final-decision distribution from papers without two-sided participation. This is an observational association, not a claim that replying causes acceptance.

Planned analysis and evaluation

We will compare against a mean-rating-only baseline at *the same prediction time*: initial mean versus models using only information available after initial reviews, and final mean versus models using information available after rebuttal but before the decision. Bayesian posterior distributions will quantify uncertainty in the three hypothesized effects and in predicted outcome probabilities. Posterior predictive checks will assess whether the model reproduces observed decision patterns; held-out multiclass log loss, calibration, and class-wise performance will assess predictive value, especially for the relatively rare spotlight and oral classes. Hypotheses may be refined using training data, but the final claims and model comparison will be evaluated on held-out papers. Since this study uses only ICLR 2025, it will assess prediction within that year rather than generalization to future conferences. Because the data are observational, estimated effects will be interpreted as predictive associations rather than causal effects.